A NOETHERIAN LOCAL RING THAT IS NOT FORMALLY CATENARY IS NOT UNIVERSALLY CATENARY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A finite type extension of the ring can be built whose completion is the quotient of the completion by a prescribed ideal that has only the maximal ideal as associated prime. invoked at proof lines 32335-32340. Produces the finite type ring B with completion A-hat modulo J in the case of dimension one.
- The dimension formula for a flat local map of Noetherian local rings, comparing dimensions with the transcendence degrees of the residue field and of the fibre. invoked at proof lines 32348-32358. Supplies the inequality that contradicts the dimension count once a prime lying over the zero ideal is produced.
- Prime avoidance lets one choose an element of the maximal ideal avoiding finitely many prescribed primes. invoked at proof lines 32378-32380. Produces the element f used in the induction step so that the quotient by f is again not formally catenary with the invariant decreased.

Source reference. Original lemma 0ALR: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32037–32037. Invoked at proof lines 32338-32340 to produce the finite type ring B whose completion is A-hat modulo a power of q; the verification that the maximal ideal is the only associated prime of that ideal, which is the hypothesis of the reference, is inside the counted span.

Source reference. Original lemma 02IJ: sources/stacks-chow/tags.txt, source lines 1–1. Applied at proof lines 32350-32360 to derive an inequality contradicting the computed dimension of the localization of B once a prime over the zero ideal is available; producing that prime and the dimension computation stay inside the counted span.

Source reference. Original lemma 0394: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 32341-32343 to identify the completion of B with the adic completion at the maximal ideal and to compute its dimension; the surrounding dimension count is inside the counted span.

Source reference. Original lemma 00DS: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 32379-32380 to pick an element of the maximal ideal outside finitely many primes; the verification that the quotiented ring is again not formally catenary with the invariant decreased is inside the counted span.

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Lemma 0.2. *Let $(A, \mathfrak{m})$ be a Noetherian local ring which is not formally catenary. Then A is not universally catenary.*

Proof. By assumption there exists a minimal prime $\mathfrak{p} \subset A$ such that the spectrum of $A^\wedge/\mathfrak{p}A^\wedge$ is not equidimensional. After replacing A by $A/\mathfrak{p}$ we may assume that A is a domain and that the spectrum of $A^\wedge$ is not equidimensional. Let $\mathfrak{q}$ be a minimal prime of $A^\wedge$ such that $d = \dim(A^\wedge/\mathfrak{q})$ is minimal and hence $0 < d < \dim(A)$. We prove the lemma by induction on d.

The case $d = 1$. In this case $\dim(A^\wedge_\mathfrak{q}) = 0$. Hence $A^\wedge_\mathfrak{q}$ is Artinian local and we see that for some $n > 0$ the ideal $J = \mathfrak{q}^n$ maps to zero in $A^\wedge_\mathfrak{q}$. It follows that $\mathfrak{m}$ is the only associated prime of J/J^2, whence $\mathfrak{m}^m$ annihilates J/J^2 for some $m > 0$. Thus we can use Lemma 0ALR to find $A \to B$ of finite type such that $B^\wedge \cong A^\wedge/J$. It follows that $\mathfrak{m}_B = \sqrt{\mathfrak{m}B}$ is a maximal ideal with the same residue field as $\mathfrak{m}$ and $B^\wedge$ is the $\mathfrak{m}_B$-adic completion (Algebra, Lemma 0394). Then

$$\dim(B_{\mathfrak{m}_B}) = \dim(B^\wedge) = 1 = d.$$

Since we have the factorization $A \to B \to A^\wedge/J$ the inverse image of $\mathfrak{q}/J$ is a prime $\mathfrak{q}' \subset \mathfrak{m}_B$ lying over (0) in A. Thus, if A were universally catenary, the dimension formula (Algebra, Lemma 02IJ) would give

$$\begin{aligned}\dim(B_{\mathfrak{m}_B}) &\geq \dim((B/\mathfrak{q}')_{\mathfrak{m}_B}) \\ &= \dim(A) + \operatorname{trdeg}_A(B/\mathfrak{q}') - \operatorname{trdeg}_{\kappa(\mathfrak{m})}(\kappa(\mathfrak{m}_B)) \\ &= \dim(A) + \operatorname{trdeg}_A(B/\mathfrak{q}')\end{aligned}$$

This contradiction finishes the argument in case $d = 1$.

Assume $d > 1$. Let $Z \subset \operatorname{Spec}(A^\wedge)$ be the union of the irreducible components distinct from $V(\mathfrak{q})$. Let $\mathfrak{r}_1, \ldots, \mathfrak{r}_m \subset A^\wedge$ be the prime ideals corresponding to irreducible components of $V(\mathfrak{q}) \cap Z$ of dimension > 0. Choose $f \in \mathfrak{m}$, $f \notin A \cap \mathfrak{r}_j$ using prime avoidance (Algebra, Lemma 00DS). Then $\dim(A/fA) = \dim(A) - 1$ and there is some irreducible component of $V(\mathfrak{q}, f)$ of dimension $d-1$. Thus A/fA is not formally catenary and the invariant d has decreased. By induction A/fA is not universally catenary, hence A is not universally catenary. $\square$